

```
% run k-means
```

```
close all
```

```
clear
```

```
clc
```

```
load workspace.mat
```

```
figure
```

```
plot(c_x, c_y, 'b*')
```

```
hold on
```

```
title('Linear Regression Prediction')
```

```
P = polyfit(c_x, c_y, 1)
```

```
p_y = polyval(P, [1:100]);
```

```
plot([1:100], p_y, 'k-')
```

```
printfig77(gcf, 'linear_regression')
```

```
figure
```

```
plot(c_x, c_y, 'b*')
```

```
hold on
```

```
mu_x = mean(c_x);
```

```
mu_y = mean(c_y);
```

```
plot([1, 100], [mu_y, mu_y], 'k-')
```

```
plot([mu_x, mu_x], [1, 100], 'k-')
```

```
title('Simple mean, ONE CLUSTER')
```

```
printfig77(gcf, 'one_cluster')
```

```
% start with two points, since the data visually has two clusters
```

```
figure
```

```
plot(c_x, c_y, 'b*')
```

```
hold on
```

```
cluster = [];
```

```
data = [c_x, c_y];
```

```
for i = 1:3
```

```
    idx = ceil(rand*size(data, 1));
```

```
    cluster{i} = data(idx, :);
```

```
    retain = ones(size(data), 1);
```

```
    retain(idx) = 0;
```

```
    data = data(retain>0, :);
```

```
    plot(cluster{i}(1), cluster{i}(2), 'ko', 'MarkerSize', 9)
```

```
end
```

```
title('Select the intial centroids')
```

```
printfig77(gcf, 'two_cluster')
```

```
% then randomly select the next data point to calculate distances to
figure
plot(data(:,1), data(:,2), 'b.')
hold on
for i = 1:size(cluster{1}, 1)
    plot(cluster{1}(i, 1), cluster{1}(i, 2), 'g.')
end
for i = 1:size(cluster{2}, 1)
    plot(cluster{2}(i, 1), cluster{2}(i, 2), 'm.')
end
for i = 1:size(cluster{3}, 1)
    plot(cluster{3}(i, 1), cluster{3}(i, 2), 'c.')
end

idx = ceil(rand*length(data));
val = data(idx, :);
for i = 1:3
    dist(i) = sqrt(sum( (val - mean(cluster{i}, 1)) .^ 2));
end

s = 1;
centroid_this = mean(cluster{s}, 1);
plot(centroid_this(1), centroid_this(2), 'g+', 'MarkerSize', 12);
plot([val(1), centroid_this(1)], [val(2), centroid_this(2)], 'r-')
s = 2;
centroid_this = mean(cluster{s}, 1);
plot(centroid_this(1), centroid_this(2), 'm+', 'MarkerSize', 12);
plot([val(1), centroid_this(1)], [val(2), centroid_this(2)], 'r-')
s = 3;
centroid_this = mean(cluster{s}, 1);
plot(centroid_this(1), centroid_this(2), 'm+', 'MarkerSize', 12);
plot([val(1), centroid_this(1)], [val(2), centroid_this(2)], 'r-')

[y, m] = min(dist); % m in the index of the minimum distance
cluster{m} = [cluster{m}; val];

% now pop this data point out of the data set
retain = ones(size(data), 1);
retain(idx) = 0;
data = data(retain>0, :);
title('First Assignment')
```

```
printfig77(gcf, 'first_assignment')
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
figure
plot(data(:,1), data(:,2), 'b.')
hold on
for i = 1:size(cluster{1}, 1)
    plot(cluster{1}(i, 1), cluster{1}(i, 2), 'g.')
end
for i = 1:size(cluster{2}, 1)
    plot(cluster{2}(i, 1), cluster{2}(i, 2), 'm.')
end

idx = ceil(rand*length(data));
val = data(idx, :);
for i = 1:2
    dist(i) = sqrt(sum( (val - mean(cluster{i}, 1)) .^ 2));
end

s = 1;
centroid_this = mean(cluster{s}, 1);
plot(centroid_this(1), centroid_this(2), 'g+', 'MarkerSize', 12);
plot([val(1), centroid_this(1)], [val(2), centroid_this(2)], 'r-')
s = 2;
centroid_this = mean(cluster{s}, 1);
plot(centroid_this(1), centroid_this(2), 'm+', 'MarkerSize', 12);
plot([val(1), centroid_this(1)], [val(2), centroid_this(2)], 'r-')
```

```
[y, m] = min(dist); % m in the index of the minimum distance
cluster{m} = [cluster{m}; val];
```

```
% now pop this data point out of the data set
retain = ones(size(data), 1);
retain(idx) = 0;
data = data(retain>0, :);
title('Second Assignment')
printfig77(gcf, 'second_assignment')
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
figure
plot(data(:,1), data(:,2), 'b.')
hold on
for i = 1:size(cluster{1}, 1)
```

```

    plot(cluster{1}(i, 1), cluster{1}(i, 2), 'g.')
end
for i = 1:size(cluster{2}, 1)
    plot(cluster{2}(i, 1), cluster{2}(i, 2), 'm.')
end

idx = ceil(rand*length(data));
val = data(idx, :);
for i = 1:2
    dist(i) = sqrt(sum( (val - mean(cluster{i}, 1)) .^ 2));
end

s = 1;
centroid_this = mean(cluster{s}, 1);
plot(centroid_this(1), centroid_this(2), 'g+', 'MarkerSize', 12);
plot([val(1), centroid_this(1)], [val(2), centroid_this(2)], 'r-')
s = 2;
centroid_this = mean(cluster{s}, 1);
plot(centroid_this(1), centroid_this(2), 'm+', 'MarkerSize', 12);
plot([val(1), centroid_this(1)], [val(2), centroid_this(2)], 'r-')

[y, m] = min(dist); % m in the index of the minimum distance
cluster{m} = [cluster{m}; val];

% now pop this data point out of the data set
retain = ones(size(data), 1);
retain(idx) = 0;
data = data(retain>0, :);
title('Third Assignment')
printfig77(gcf, 'third_assignment')

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

figure

while(length(data) > 0)
    if (mod(size(data, 1), 10) == 0)
        figurtime = true;
    else
        figurtime = false;
    end

    if (figurtime)
        plot(data(:,1), data(:,2), 'b.')
        hold on

```

```

    for i = 1:size(cluster{1}, 1)
        plot(cluster{1}(i, 1), cluster{1}(i, 2), 'g.')
    end
    for i = 1:size(cluster{2}, 1)
        plot(cluster{2}(i, 1), cluster{2}(i, 2), 'm.')
    end
end

idx = ceil(rand*length(data));
val = data(idx, :);
for i = 1:2
    dist(i) = sqrt(sum( (val - mean(cluster{i}, 1)) .^ 2));
end

if (figurtime)
    s = 1;
    centroid_this = mean(cluster{s}, 1);
    plot(centroid_this(1), centroid_this(2), 'g+', 'MarkerSize', 12);
    plot([val(1), centroid_this(1)], [val(2), centroid_this(2)], 'r-')
    s = 2;
    centroid_this = mean(cluster{s}, 1);
    plot(centroid_this(1), centroid_this(2), 'm+', 'MarkerSize', 12);
    plot([val(1), centroid_this(1)], [val(2), centroid_this(2)], 'r-')
end

[y, m] = min(dist); % m in the index of the minimum distance
cluster{m} = [cluster{m}; val];

% now pop this data point out of the data set
retain = ones(size(data), 1);
retain(idx) = 0;
data = data(retain>0, :);
hold off
if (figurtime)
    title([num2str(size(data, 1)), ' remaining'])
    printfig77(gcf, ['xstep_', num2str(size(data, 1))])
end
end
end

```

```
%%%%%%%%%
```

```
figure
```

```

s = 1;
centroid_this = mean(cluster{s}, 1);
plot(centroid_this(1), centroid_this(2), 'k+', 'MarkerSize', 12);

```

```
    hold on  
s = 2;  
    centroid_this = mean(cluster{s}, 1);  
    plot(centroid_this(1), centroid_this(2), 'k+', 'MarkerSize', 12);  
  
for i = 1:size(cluster{1}, 1)  
    plot(cluster{1}(i, 1), cluster{1}(i, 2), 'g.')  
end  
for i = 1:size(cluster{2}, 1)  
    plot(cluster{2}(i, 1), cluster{2}(i, 2), 'm.')  
end  
title('Finished K-Means')  
printfig77(gcf, 'kmeans_done')
```